of personal audio devices. Join us as we delve into the basics of the electronic engineering that goes into audio technology to find out what it all means.

Analog Signals - The structure of Sound

Let us begin with a blank slate and work our way back from the conscious perception of sound. On second thought, let us just work our way back from the moment sound hits our ears.

At the risk of turning it into a platitude within this FastTrack, Sound is a wave. As we've seen, waves need a medium to travel in. When you hear anything, the waves are variations in air pressure, which mechanically vibrate your eardrums. These air waves are created by the instruments (if you are listening to an acoustic performance), or by the vibrating membrane of the speaker. The speaker cone produces the desired sound because it is being fed the desired electrical signal, whose variations conform precisely with the curves of the sound wave. We won't go into detail as this has been covered in Chapter 2. The point to be illustrated here is, the signal that goes into the speaker is an analog audio signal. Prior to that, the signal may have been analog or digital, and then passed through a DAC, which we will examine shortly. To record an analog signal from a source of sound, we need to arrange apparatus such that the instantaneous change in air pressure is directly proportional to the instantaneous change in some property of the physical media. Some years ago that was vinyl discs and magnetic tapes. The pattern of continuous change over time that exactly corresponds to the structure of the sound is the analog recording.

In today's scenario the physical media is electricity and the varying property is voltage, or sometimes current. In this case, we cannot store analog electrical signals as there are various problems associated with this idea, which we will see as we go digital.

The Digital Discourse

The brief history of today's electronic technology begins with the humble three terminal transistor. The discernible ON and OFF states of transistors led to logic gates which led to flip flops which led to registers and processors and combined with various types of sensors and actuators, resulted in the world of bits we have today. The essence of the world of bits is that everything is dualistic at its core. Something or nothing, 1 or 0, the fundamental digits. From that humble beginning we have built systems as complex as climate simulation using increasingly higher levels of